



Introduction to Spring and DI

1. Introduction to spring
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- Spring Framework is an application framework.
- It's primary task is to make it easier to build Java applications.
- It handles common tasks that developers often have to deal with, like connecting to databases, managing transactions, and handling web requests etc.
- Consider spring framework like a tool box having pre built tools
- This framework uses various new techniques such as Aspect-Oriented Programming (AOP), Plain Old Java Object (POJO), and dependency injection (DI)
- Development of Web applications becomes much easier as compared to classic Java frameworks and Application Programming Interfaces (APIs), such as Java database connectivity(JDBC), JavaServer Pages(JSP), and Java Servlet.

- It solves difficulties of Enterprise application development
- Support Enterprise application development through POJOs
- Easy integration with other frameworks
- Application Testing is much easier
- Spring modules can be used as per application requirement in modular fashion
- Spring Transaction Management interface is very flexible
- It supports both XML- and annotation-based configuration

- Working with Spring is more complex. The learning curve is also difficult.
- It provides multiple options to developers that may create confusion of developers.
- Developing a Spring application requires lots of XML initially.

- The Spring Framework consists of features organized into different modules which are grouped.
- Core Container Layer:
 - Core module contains the framework's core components, such as the IoC and Dependency Injection capabilities.
 - Context module : provides a means for accessing any objects established and configured
- Data Access/Integration Layer:
 - The JDBC module provides a JDBC-abstraction layer that removes the need to do tedious JDBC coding and parsing of database-vendor specific error codes.
 - The ORM module provides integration layers for popular object-relational mapping APIs, including Hibernate.

- Web layer:
 - Spring's MVC framework provides a clean separation between domain model code and web forms
- AOP and Instrumentation layer:
 - AOP allows you to separate concerns that cut across multiple modules or objects into distinct modules called aspects.
- Test layer
 - Supports the testing of Spring components with JUnit or TestNG

- In traditional programming, objects that are statically assigned to one another.
- IOC (Inversion of Control) refers to the principle of transferring the responsibility of creating and managing dependencies to an external source.
- IOC is a broader concept about the inversion of control of dependencies

- DI (Dependency Injection) is a specific technique for implementing IOC by injecting dependencies.
- DI needs to be done by an external entity instead of creating the dependencies by the component itself.
- Dependency injection is a process of injecting (pushing) the dependencies into an object.
- Traditional ways of obtaining the dependencies are by creating the dependencies or by pulling the dependencies using some factory classes or methods or from the naming registry.